

The Health Dividend – Initial Ideas

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Motivating Questions: every year governments spend billions of dollars on healthcare, but how do we actually know if it's working? Does a country that allocates more of their GDP to health initiatives actually have healthier people, lower mortality rates, or longer lives relative to the global standard?

Our goal is to use [this Global Health Indicators Dataset](#) to query, compare, and analyze the following variables over the years 2015-2019:

- **Current Health Expenditure (% of GDP):** Shows the percentage of a country's GDP allocated to health expenditure.
- **Life Expectancy at Birth (Total Years):** The average number of years a newborn is expected to live, assuming age-specific mortality rates remain constant.
- **Maternal Mortality:** The number of maternal deaths per 100,000 live births.
- **Infant Mortality Rate:** The number of infant deaths (under 1 year) per 1,000 live births.
- **Neonatal Mortality Rate:** The number of deaths of children under 28 days of age per 1,000 live births.
- **Under-5 Mortality Rate:** The number of deaths of children under 5 years of age per 1,000 live births.
- **Prevalence of HIV (% of population):** The percentage of the population aged 15-49 years living with HIV.
- **Incidence of Tuberculosis (per 100,000 people):** The number of incidence of tuberculosis per 100,000 people.
- **Prevalence of Undernourishment (% of population):** The percentage of the population whose caloric intake is below the minimum required for a healthy life.

See an example of some data below. Every year in the 1999-2023 range has data for the same, large set of countries. The column headers are in order of the bullets above:

i	Ireland	IRL	2016	7.46859169	81.65366	5	3.1	2.3	3.6	0.2	7.1	2.5
i	Iran, Islam	IRN	2016	7.99202633	75.6	20	12.8	9.6	14.9	0.1	15	7.2
i	Iraq	IRQ	2016	4.58949184	68.988	85	23.9	16.4	28.6	0.1	35	16.6
i	Iceland	ISL	2016	8.1	82.20488	3	2.1	1.3	2.6	0.1	2.1	2.5
i	Israel	ISR	2016	7.1	82.40732	3	3.1	2	3.8		3.5	2.5
i	Italy	ITA	2016	8.73	83.2439	6	2.9	2.1	3.4	0.3	7.2	2.5
i	Jamaica	JAM	2016	5.70102215	72.02	98	16	14.1	18.6	1.3	4.4	7.6
i	Jordan	JOR	2016	6.97644138	75.215	43	14.6	10	17	0.1	5.4	7.8
i	Japan	JPN	2016	10.66	83.98488	5	2	0.9	2.7		16	2.7
i	Kazakhsta	KAZ	2016	3.42	72.41	13	9.8	5	11		73	2.5
i	Kenya	KEN	2016	4.75429106	62.163	505	33.6	21.9	46.3	4.8	348	21.8
i	Kyrgyz Rep	KGZ	2016	6.41	70.95122	53	18.8	14.2	21	0.2	133	5.7
i	Cambodia	KHM	2016	6.11648464	70.224	197	25.9	15.5	30.1	0.7	348	7
i	Kiribati	KIR	2016	10.47	66.332	117	46.1	23.8	59.6		547	2.9
i	St. Kitts an	KNA	2016	5.42126036	71.74		15.5	12	18		0	
i	Korea, Rep	KOR	2016	6.87049913	82.27561	6	2.9	1.6	3.4		76	2.5
i	Kuwait	KWT	2016	4.73096323	79.701	6	7.7	4.9	9	0.1	24	2.5
i	Latin Amer	LAC	2016	7.870123386	74.47154	73	15.8297	10.07167	18.5896	0.4692859	43	5.259134
i	Lao PDR	LAO	2016	2.36087489	67.167	167	40	23.7	50.8	0.3	175	7.1
i	Lebanon	LBN	2016	7.92386055	79.505	19	11.2	7.7	13	0.1	12	5.8
i	Liberia	LBR	2016	9.90062618	60.416	628	63.1	32.5	86	1.2	308	34.9
i	Libya	LBY	2016	71.756	74	10.9	6.8	12.8		0.2	40	8
i	St. Lucia	LCA	2016	4.44803333	73.112	64	16.2	12.1	17.7		2	
i	Latin Amer	LCN	2016	7.890377615	74.58099	76	15.9	10.2	18.6	0.475727401	42	5.784749
i	Least deve	LDC	2016	4.177321697	64.01163	404	49.29661	28.27679	70.65683	1.883091025		20.1702
i	Low incom	LIC	2016	5.560440963	62.23504	459	53.1	29.6	77.4	1.991365782	198	25.36238
i	Liechtenst	LIE	2016		82.25854							
i	Sri Lanka	LKA	2016	3.62	75.331	30	7.1	4.8	8.3	0.1	65	4.3
i	Lower mid	LMC	2016	3.876967729	67.96769	282	40.2	25.5	53.3	0.665039502	230	11.43548
i	Low & mid	LMY	2016	5.256420208	70.65373	240	34.14024	20.89395	46.03756	1.037734698	169	9.3858
i	Lesotho	LSO	2016	9.24261093	52.264	668	63	38.1	77.8	23.7	721	
i	Late-demc	LTE	2016	5.545603593	76.2572	31	9.396242	6.0455	11.59765			3.500814

Setting a timeline and dealing with missing data: Due to some years missing pieces of data, and the vast scale of the set, we'd like to frame our project over the years 2015-2019, instead of analyzing the entire 1999-2023 scale. These years have fuller data relative to others, and they hold for other extraneous variables like the COVID-19 pandemic in 2020.

As we move objects into our Pydantic model, this should help us sift through objects that are missing several fields (for example, see Liechtenstein above). If only one field is missing in an otherwise full Country-Year record, we may consider interpolating a datapoint, but we'd like to work with the dataset more before we decide how to proceed.

Proposed API Routes:

Route	Method	Return
/data	POST, GET, DELETE	Put data into Redis, Return all data from Redis, Delete data from Redis
/countries	GET	All records
/countries/<id>	GET	All years for one country
/countries/<id>/year	GET	One Country-Year record
/years/<year>	GET	All countries for one year

We hope this dataset will help us not only learn more about software design (data validation, containerization, Kubernetes, etc.), but also global economic policy and health patterns. Please let us know any feedback you have.